

STUDENT PORTAL PRO

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REGISTRATION:BIT/2024/74067

COURSE:BIT(Bachelors of science information technology)

UNIT:MOBILE APPLICATION DEVELOPMENT

PROJECT:STUDENT PORTAL PRO

1. INTRODUCTION

Student Portal Pro is an Android mobile application developed using Java, XML, and SQLite in Android Studio. The application allows students to register, login, view profile information, and access results through a mobile interface.

WEEK 1: PROJECT SETUP & ENVIRONMENT CONFIGURATION

Objectives

- Install and configure Android Studio
- Create Student Portal Pro project
- Set up emulator
- Understand Android project structure

explanation

I installed android studio on my computer. I then created a new project using the **Empty Views Activity** template so I could build the design from scratch.

The project settings used were:

- **Language:** Java
- **Emulator:** I used my physical phone so the application could download direct to phone

Also checked on:

- AndroidManifest.xml stores the app settings and permissions.
- The java/ folder holds the code for how the app works.
- The res/layout/ folder holds the XML files for how the app looks.

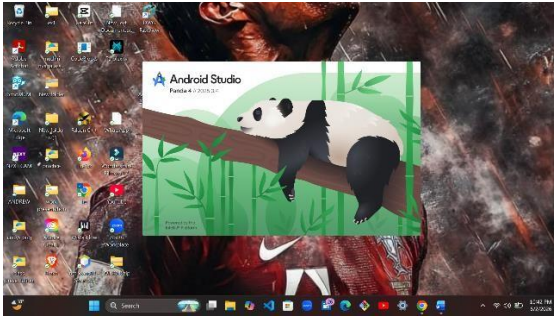


Figure 1 : ANDROID STUDIO HOMESCREEN

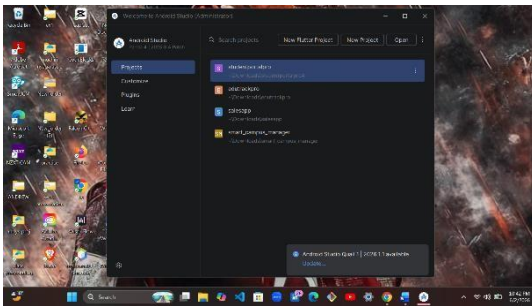


Figure 2: PROJECT CREATION SCREEN

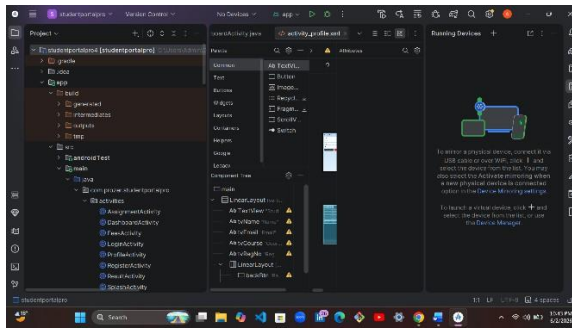


Figure 3:PROJECT STRUCTURE

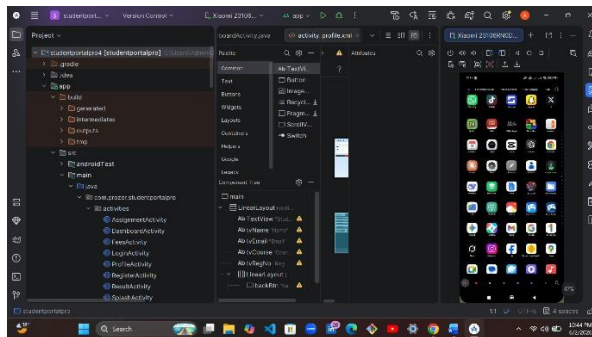


Figure 4:EMULATOR RUNNING APP

WEEK 2: UI DESIGN & ACTIVITY CREATION

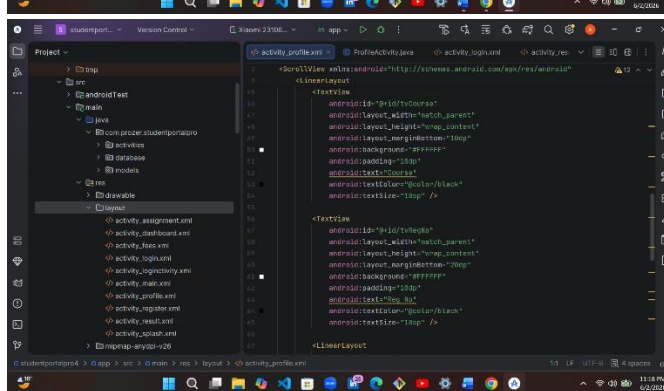
Objectives

- Create Login and Register screens
- Design UI using XML
- Implement navigation between activities
- Create Dashboard screen

Week 2 explanation

I designed the app screens using XML layouts. I used `Constraint Layout` and `Linear Layout` to make sure the text boxes and buttons look good on different phone screens.

I used `Text Input Layout` to make the input fields look neat. To move from the Login page to the Registration page, I used `Android Intents`. Once a user successfully logs in, the app takes them straight to the main Dashboard.



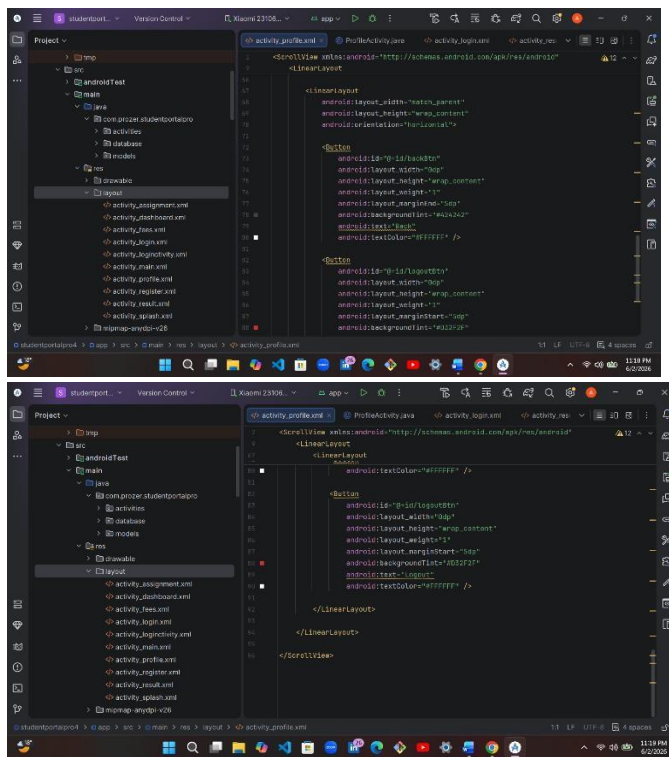
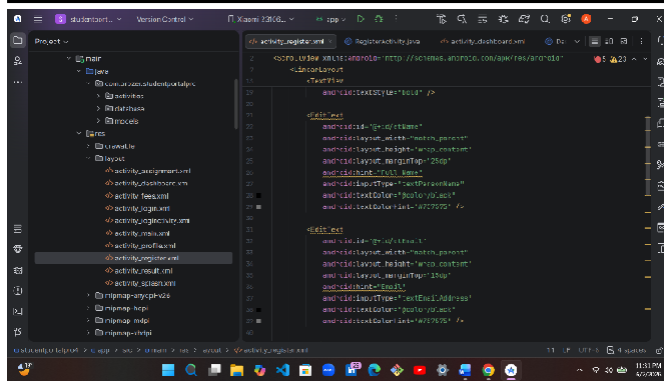
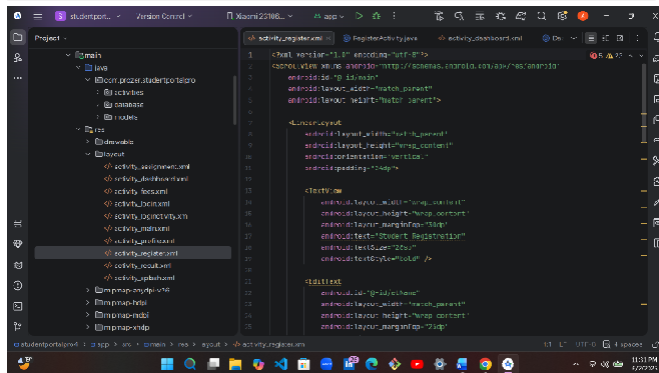
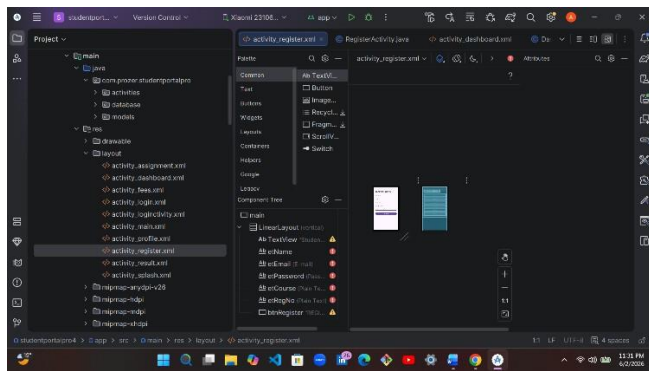


Figure 5: CODE VIEW AND ACTIVITY_LOGIN.xml code

REGISTRATION SCREENS



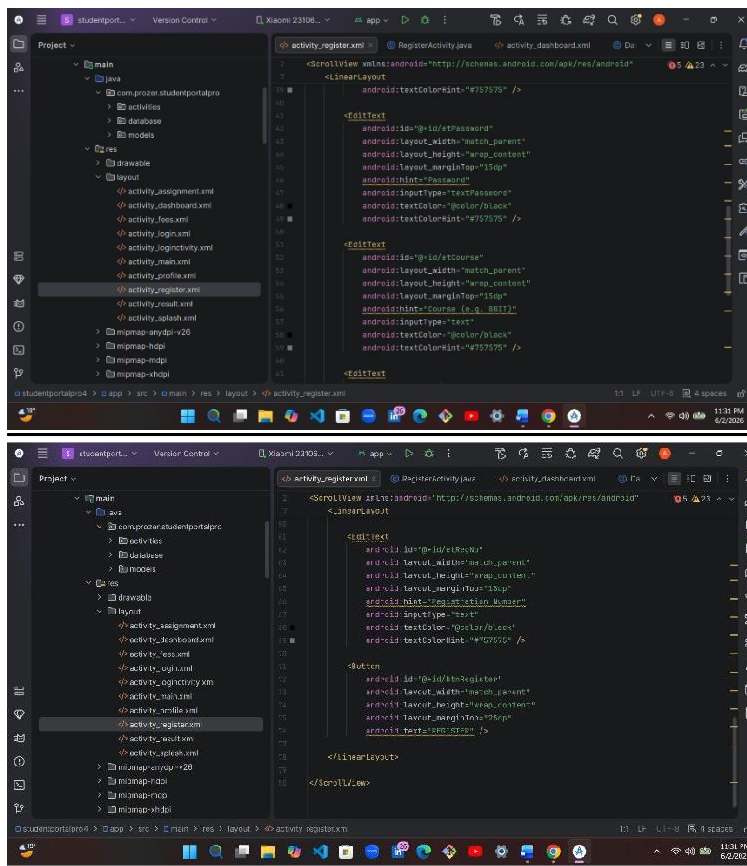


Figure 6. CODE VIEW ANDACTIVITY_REGISTER.xml

DASHBOARD SCREENS

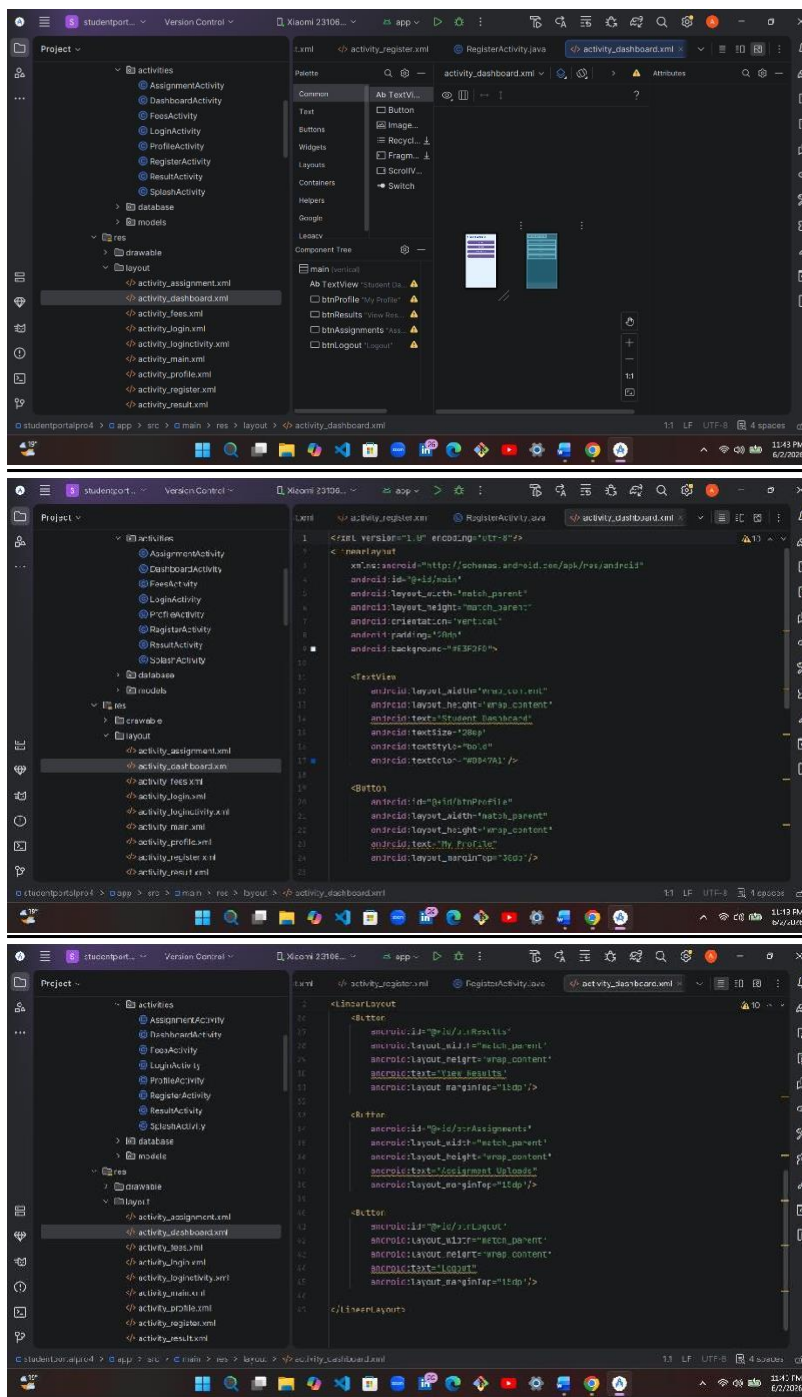


Figure 7 CODE VIEW AND ACTIVITY_DASHBOARD.XML

PROJECT STRUCTURE.

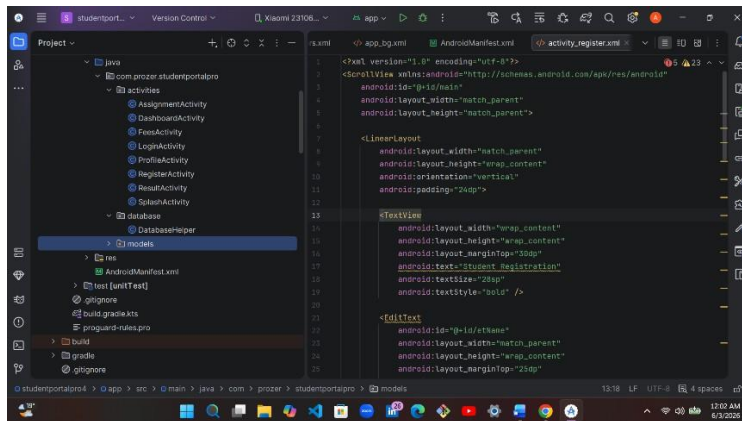


Figure 8:PROJECT STTRUCTURE

DATABASE METHODS

File: DatabaseHelper.java

REGISTER METHOD

```
Public boolean register Student(String name, String email, String password) {
```

```
    SQLite Database db = this.getWritableDatabase();  
    ContentValues values = new Content Values();
```

```
    values.put("name", name);  
    values.put("email", email);  
    values.put("password", password);
```

```
    long result = db.insert("students", null, values);
```

```
    return result != -1;  
}
```

LOGIN METHOD

```
public boolean loginStudent(String email, String password) {
```

```
    SQLiteDatabase db = this.getReadableDatabase();
```

```
    Cursor cursor = db.rawQuery(  
        "SELECT * FROM students WHERE email=? AND password=?",
```

```

        new String[]{email, password}
    );

    return cursor.getCount() > 0;
}

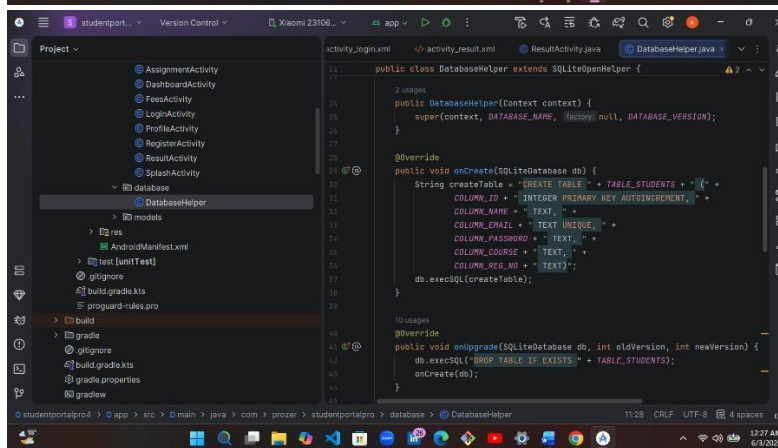
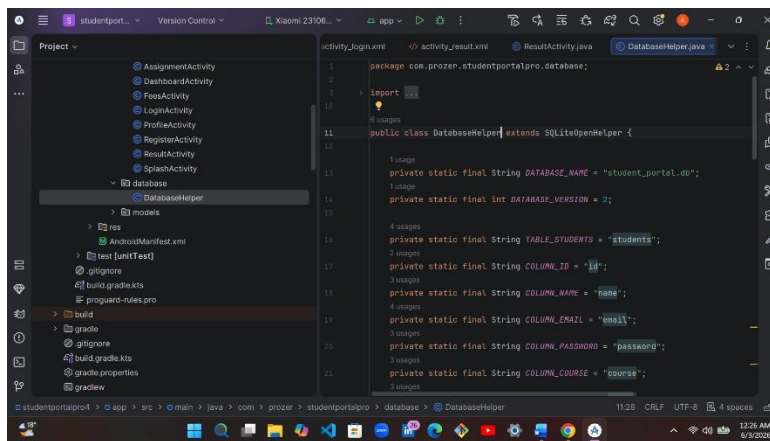
```

WEEK 3: DATABASE & FUNCTIONALITY

Objectives

- Implement SQLite database
- Enable login authentication
- Store and retrieve student data
- Test full application

DATABASE HELPER



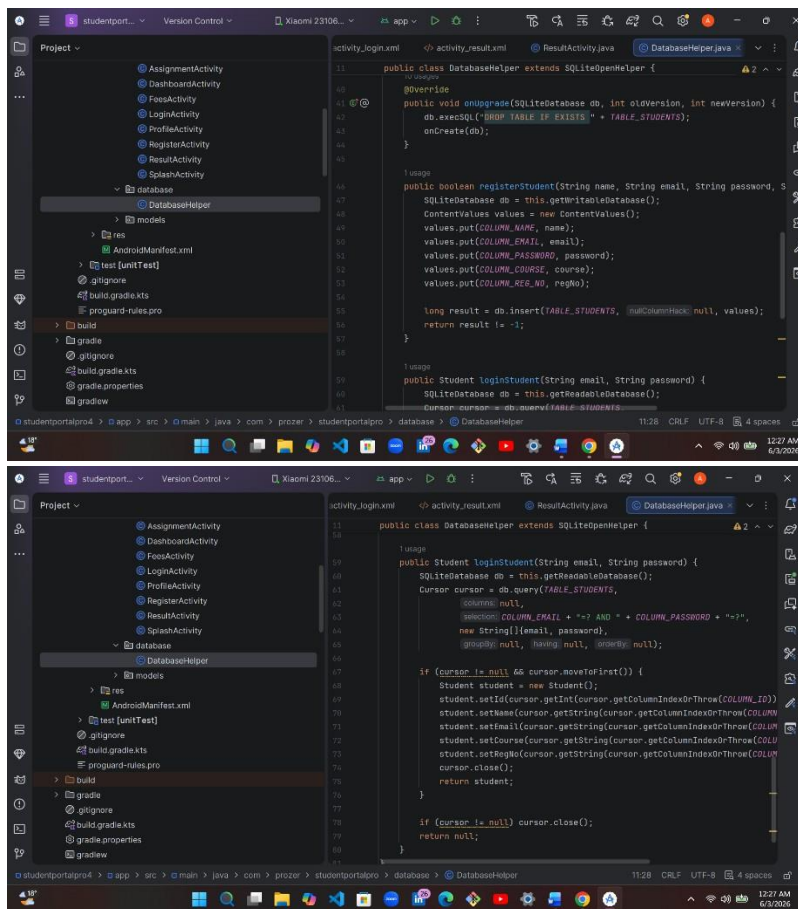
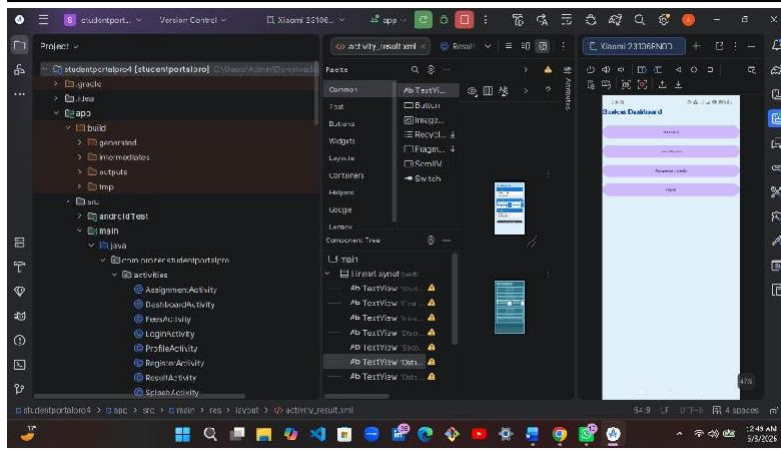
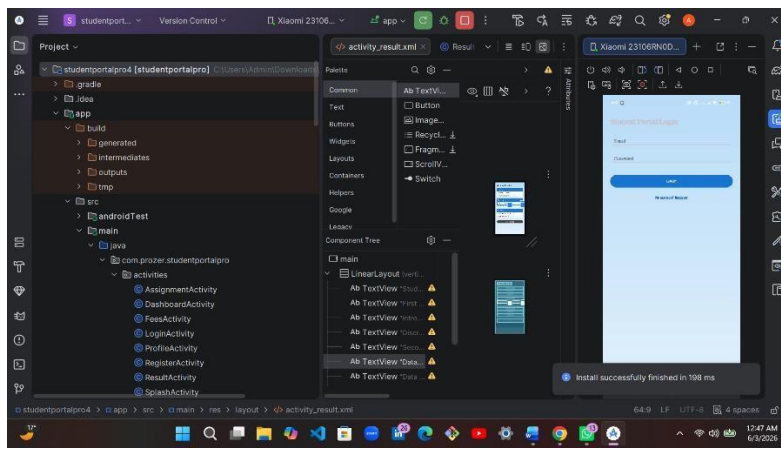


Figure 9: DATABASEHELPER

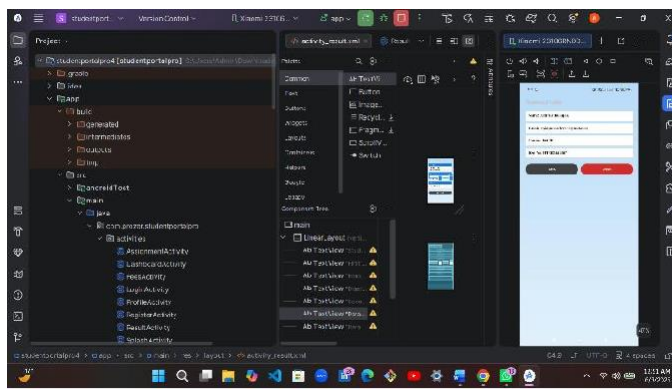
Database Code: DatabaseHelper.java

I created a class called Database Helper that extends `SQLiteOpenHelper`. This class automatically creates the database and tables when the app runs for the first time.

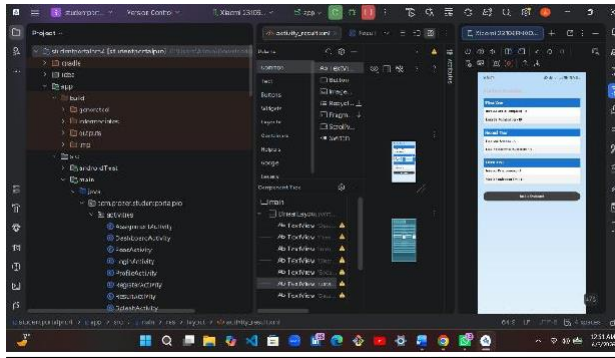
LOGIN SUCCESS SCREEN.



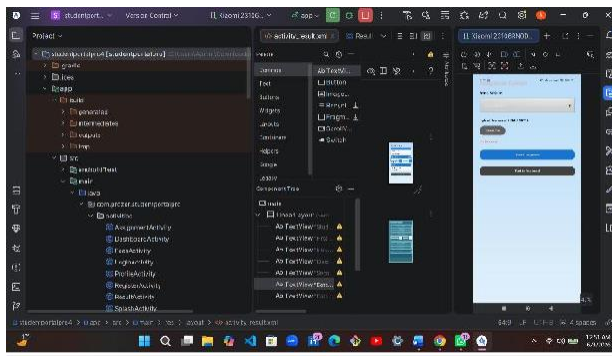
PROFILE SCREENS



RESULT SCREEN



ASSIGNMENT SCREEN UPLOADS



CONCLUSION

The_Student Portal Pro application was successfully developed using Android Studio, Java, XML, and SQLite database. The application is fully functional and allows students to register, login using valid credentials, access the dashboard, view profile information, and view results successfully. The project was tested on the Android emulator, and all major functionalities operated correctly without errors.